

Algebra 1

Unit 8

Lesson 7 Homework

Name **Answer Key** _____

Date _____

Period _____

For the expression, the first step to completing the square is shown.

$$\begin{aligned}x^2 - 8x + 15 \\(x^2 - 8x) + 15\end{aligned}$$

1. Complete the perfect square trinomial.

$$(x^2 - 8x + 16) + 15 - 16$$

$$(x - 4)^2 - 1$$

2. In the expression, the value that was added to create the perfect square trinomial is being multiplied by 1, so the value that should be subtracted from 15 to maintain equivalency is 16.

Complete the table to complete the square without using algebra tiles.

	Process	Equation
	Original equation	$y = -8x^2 - 16x + 42$
	Factor out the common factor of the first two terms.	$y = -8(x^2 + 2x) + 42$
3.	Determine the value that makes a perfect square trinomial in the parentheses.	$y = (x^2 + 2x + 1) + 42$
4.	Then, to maintain equivalence, multiply the factor by the value. Add the opposite of that value.	$y = -8(x^2 + 2x + 1) + 42 + 8$
5.	Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = -8(x + 1)^2 + 42 + 8$
6.	Combine like terms.	$y = -8(x + 1)^2 + 50$

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

7. $y = x^2 - 10x - 24$

$y = (x - 5)^2 - 49$

$y = (x - 12)(x + 2)$

8. Key Features

Key Features					
x-intercept	(12,0)	vertex	(5, -49)	domain	All Real
x-intercept	(-2,0)	increasing	$x > 5$	range	$y \geq -49$
y-intercept	(0, -24)	decreasing	$x < 5$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\infty}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\infty}$			

9. $y = 6x^2 + 12x - 48$

$y = 6(x^2 + 2x + 1) - 48 - 6$

$y = 6(x + 1)^2 - 54$

10. Key Features

Key Features					
x-intercept	(-4,0)	vertex	(-1, -54)	domain	All Real
x-intercept	(2,0)	increasing	$x < -1$	range	$y \geq -54$
y-intercept	(0, -48)	decreasing	$x > -1$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\infty}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\infty}$			